



Gen AI and Agentic AI in Production

Week 4
Day 5

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

Amazon
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

Amazon
Bedrock

Terraform

GitHub
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

Enterprise Grade



Scalability

Horizontally scaling to meet demand with no downtime



Security

Best practice information security



Monitoring

Best of breed tooling for tracking activities



Guardrails

Check LLM inputs and outputs to protect against LLM weaknesses



Explainability

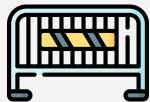
Understand the rationale for LLM outputs



Observability

Deep insight into LLM behavior as traces and spans

The Three Amigos



Guardrails

Code-based: did charter create pure json only; test for problematic language
LLM as a judge: check output coherence and alignment



Monitoring

Log errors and use alerting for anomalies



Observability

Record events and scores, track trends



Security

If you can't see it, you can't secure it

Alex - the financial planner



What it is:

SaaS commercial application, financial planner

Done:

Researcher Agent with Data Ingest

Day 1: Database

Day 2: Five Agents

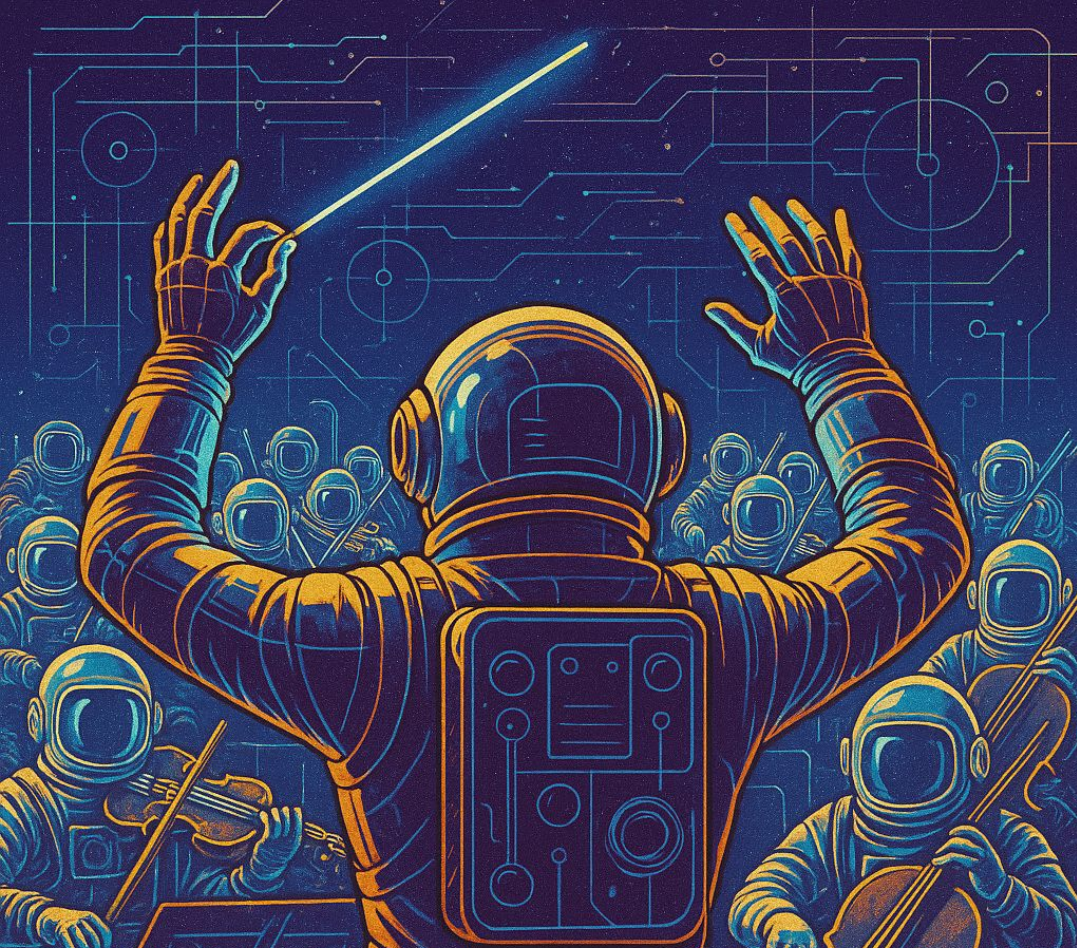
Day 3: Frontend + API

Day 4: Enterprise Grade

Todo:

Make it your own!

But for now: **terraform destroy**



AND NOW.. THE GRAND FINALE

The Agent Platform

An integrated solution for building and deploying AI Agents in managed infrastructure

Examples:

CrewAI Enterprise

LangGraph Platform

Vertex AI Agent Builder / Agent Engine

and new to the party:

Amazon Bedrock AgentCore

Pros



Fast time to market

Hugely simplified

Batteries included

Cons

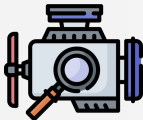


Immature market, unclear leaders

Less flexibility

Locked into ecosystem and abstractions

Amazon Bedrock AgentCore



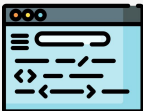
AgentCore (Preview) - replaces Bedrock Agents

A package of 5 modular AWS Services and 2 managed tools for Agentic AI



AgentCore SDK

Python runtime library to quickly build Agents for AgentCore with any common underlying Agent Framework



AgentCore Starter Toolkit

A CLI that scaffolds, deploys and operates your Agent on AgentCore

Pricing: <https://aws.amazon.com/bedrock/agentcore/pricing/>

And..

Strands Agents is Amazon's Open Source Agent Framework that is very similar to OpenAI Agents SDK and Google ADK - lightweight and convenient.

AgentCore Services & Tools

SERVICES

Runtime

Identity

Memory

Gateway

Observability

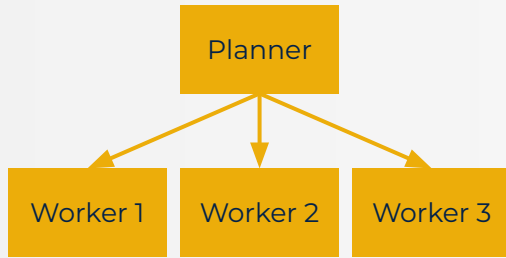
TOOLS

Browser

Code
Interpreter

Agent Architectures

Multi-Agent



Single Agent with Loop



TO THE FINAL LAB!



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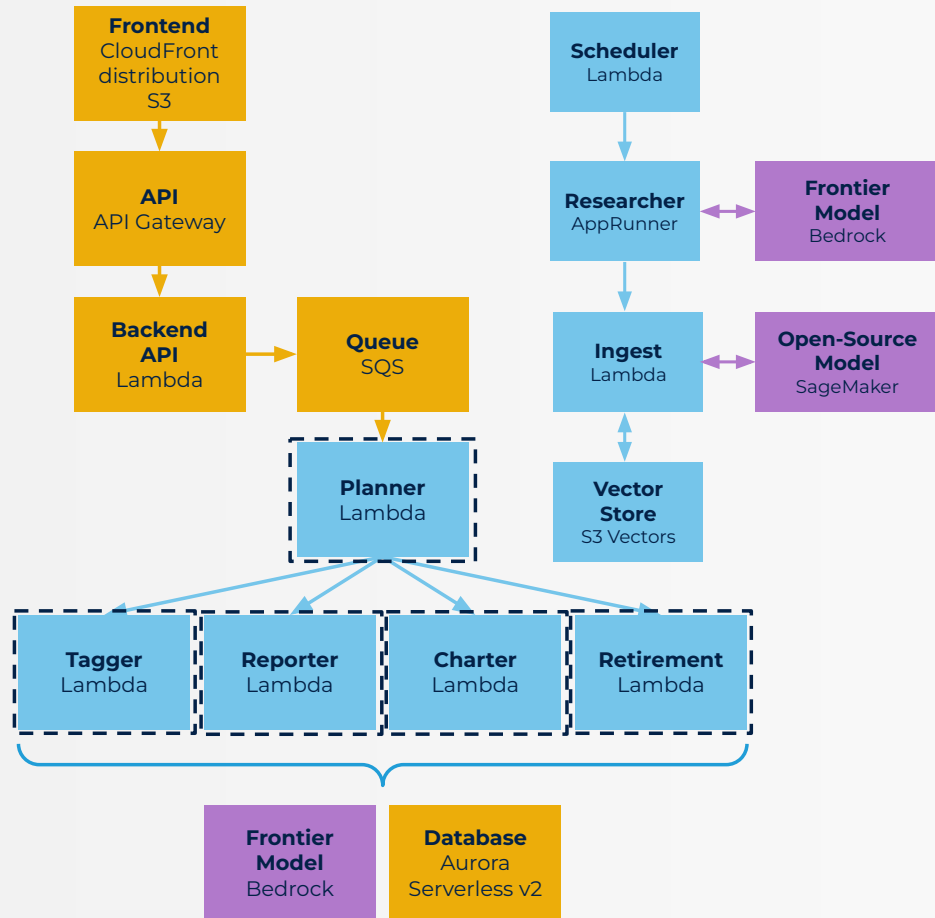
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Pin this up on your wall!!



CONGRATULATIONS

on your production expertise



Remember - I come with the package!
<https://www.linkedin.com/in/eddonner/>

Post your certificate on LinkedIn and I will amplify

Post your projects and make PRs to
community-contributions

And most importantly: **deploy to production!**



Gen AI and Agentic AI in Production

100%
Complete